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Distangling contaminants from true intratumoural microbial signals in pancreatic ductal adenocarcinoma – A benchmarking approach

Linh Dang, Johannes Richter, Louisa Eskelson, Areej Shahbaz, Jacob Hamm, Tim Beißbarth, Volker Ellenrieder, Albrecht Neesse, Christoph Ammer-Herrmenau

AG Neeße - Clinic for Gastroenterology, Gastrointestinal Oncology and Endocrinology

University Medical Center Göttingen

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Intratumoral Microbiome

Background

- Malignant tumors harbor their own bacteria community,
- New biomarkers for diagnoses, prognoses and therapy.

Challenges!!

- Low bio-mass samples, prone to contamination,
- Prone to batch effect.

→ **Need a systematic study**

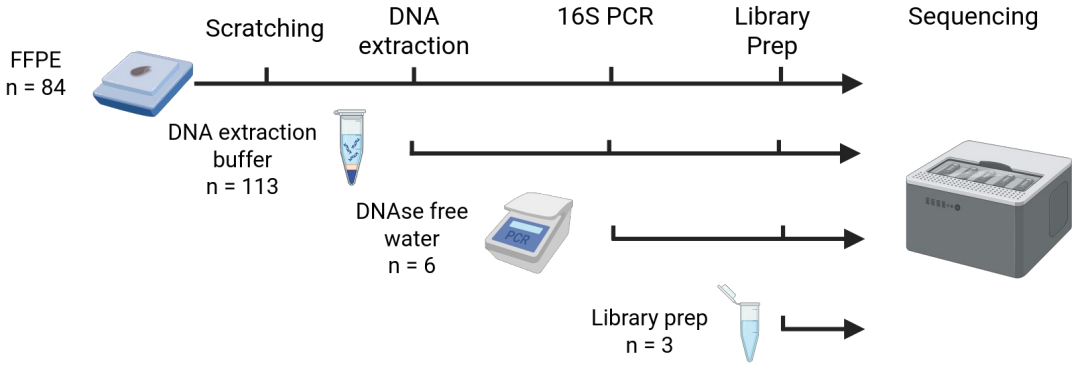
Section 1

Negative Control Survey in the Lab

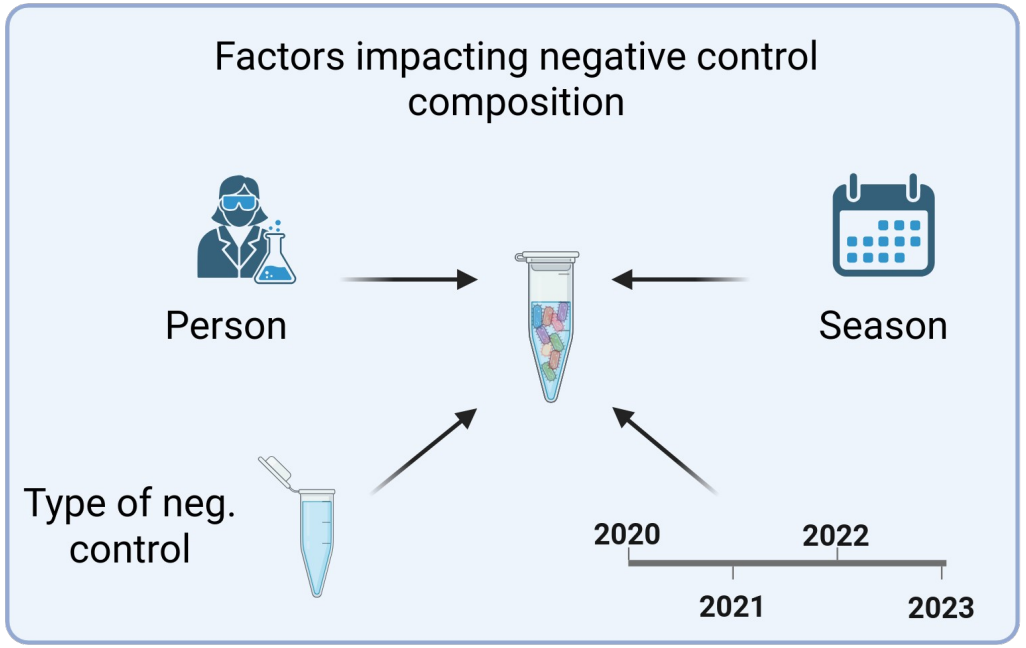
Negative Sample Collection

203 negative control samples were collected over 4 years:

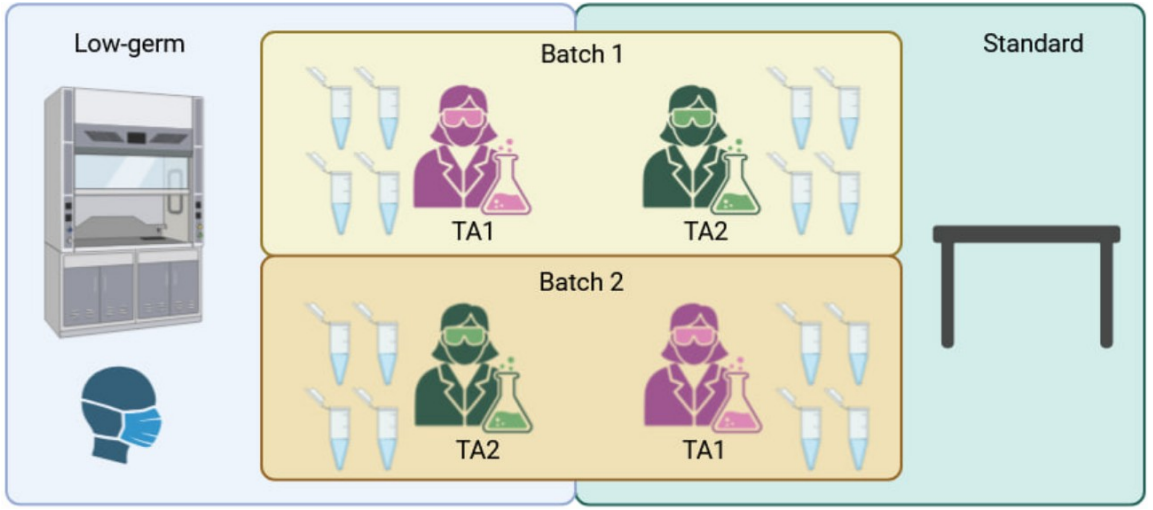
- **113** buffer controls,
- **84** paraffin controls,
- **6** PCR controls.

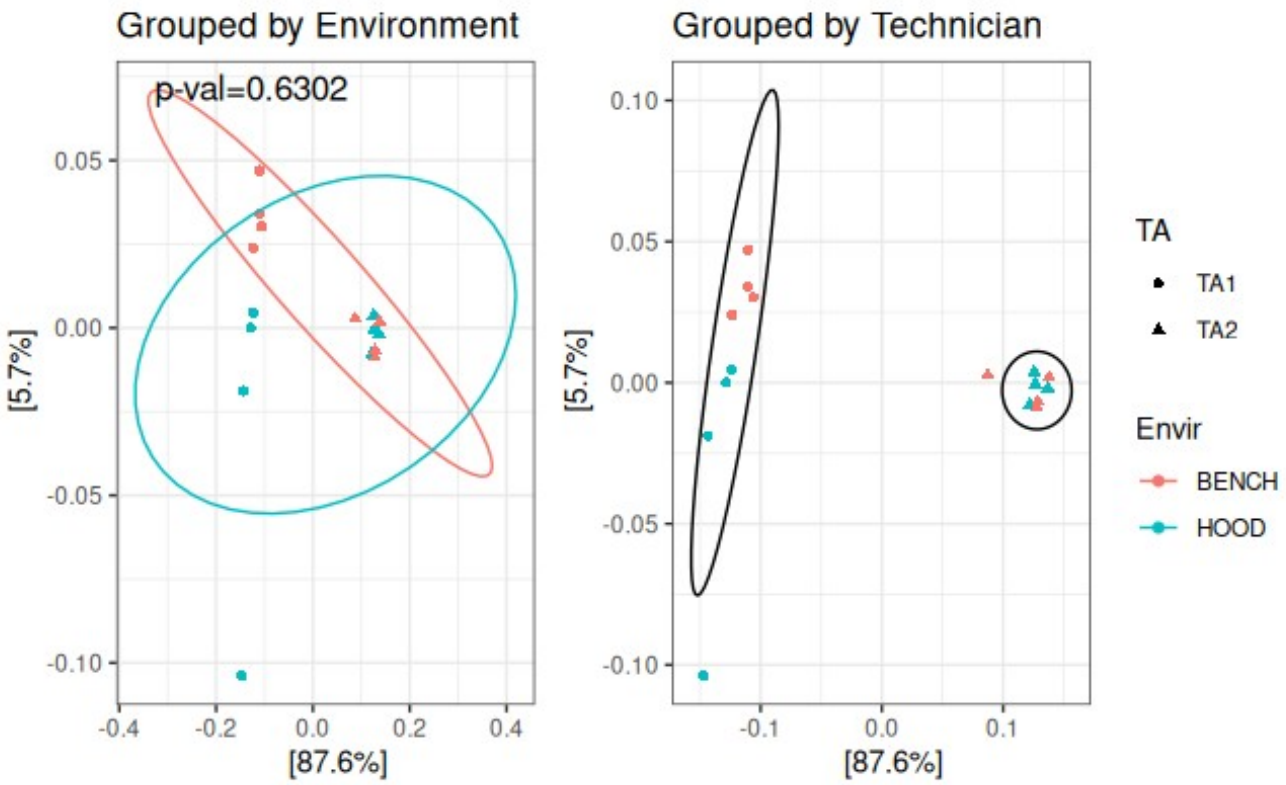


Significant batch effects!



Technician Influences Microbiome More Than Processing Environment





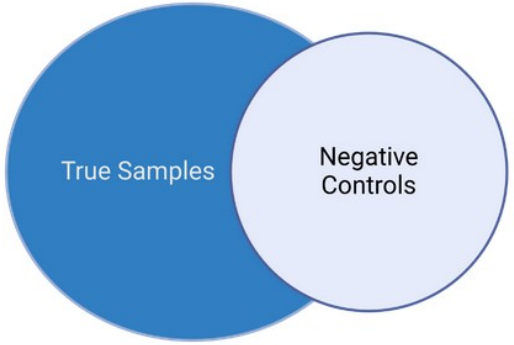
Left: created by Biorender. Right: unpublished data

Section 2

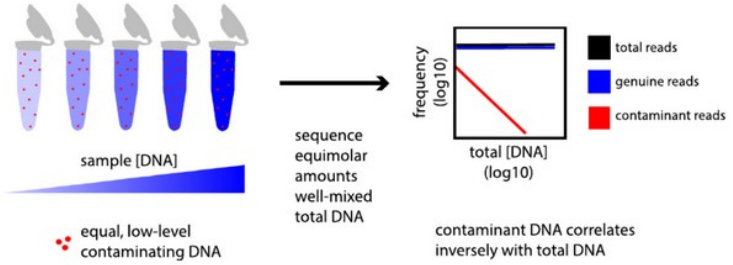
Assessment of Decontamination Methods and Intratumoral Microbiome in PDAC

Decontamination Strategies

Restrictive

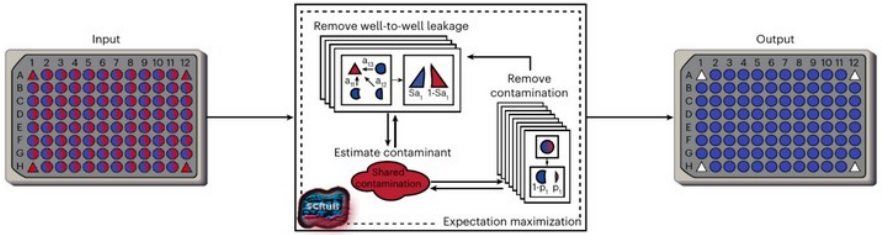


Decontam

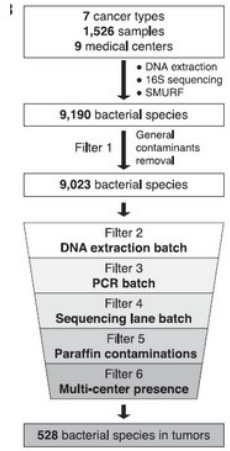


Davis et al. 2018 Microbiome

SCRuB



Austin et al. 2023 Nature Biotechnology



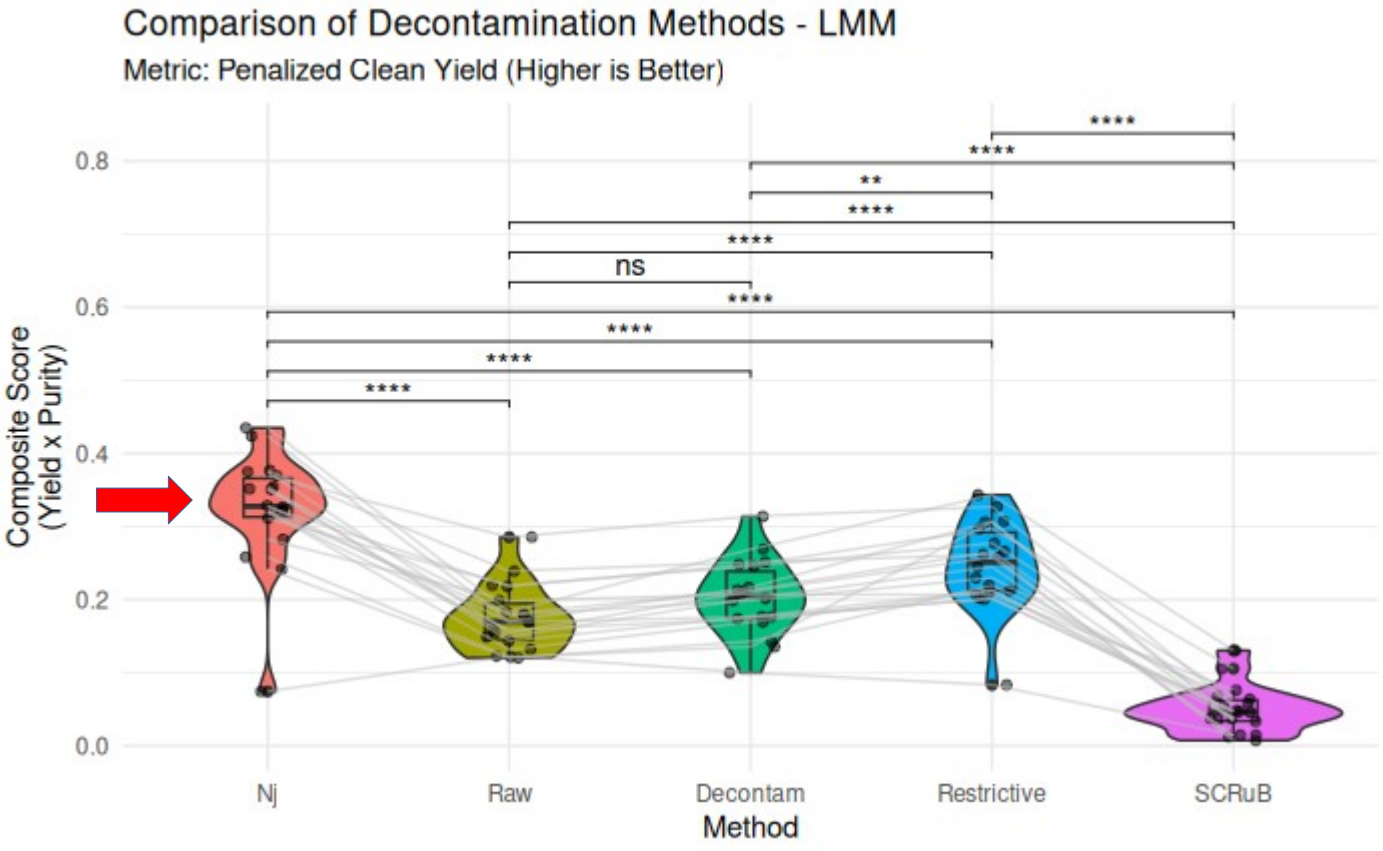
Nejman et al. 2020 Science

Assessment of Decontamination Approach

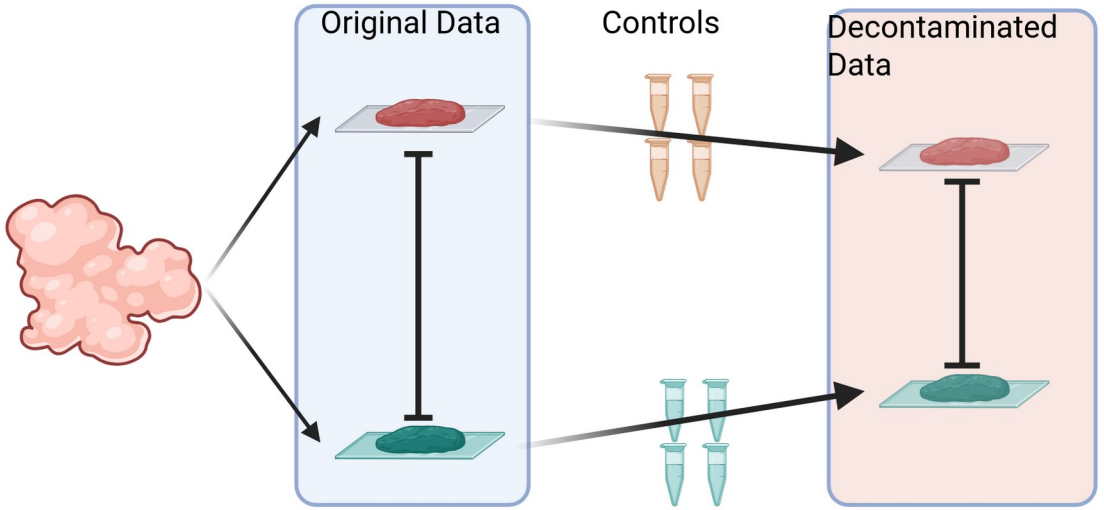
Composite Score ~

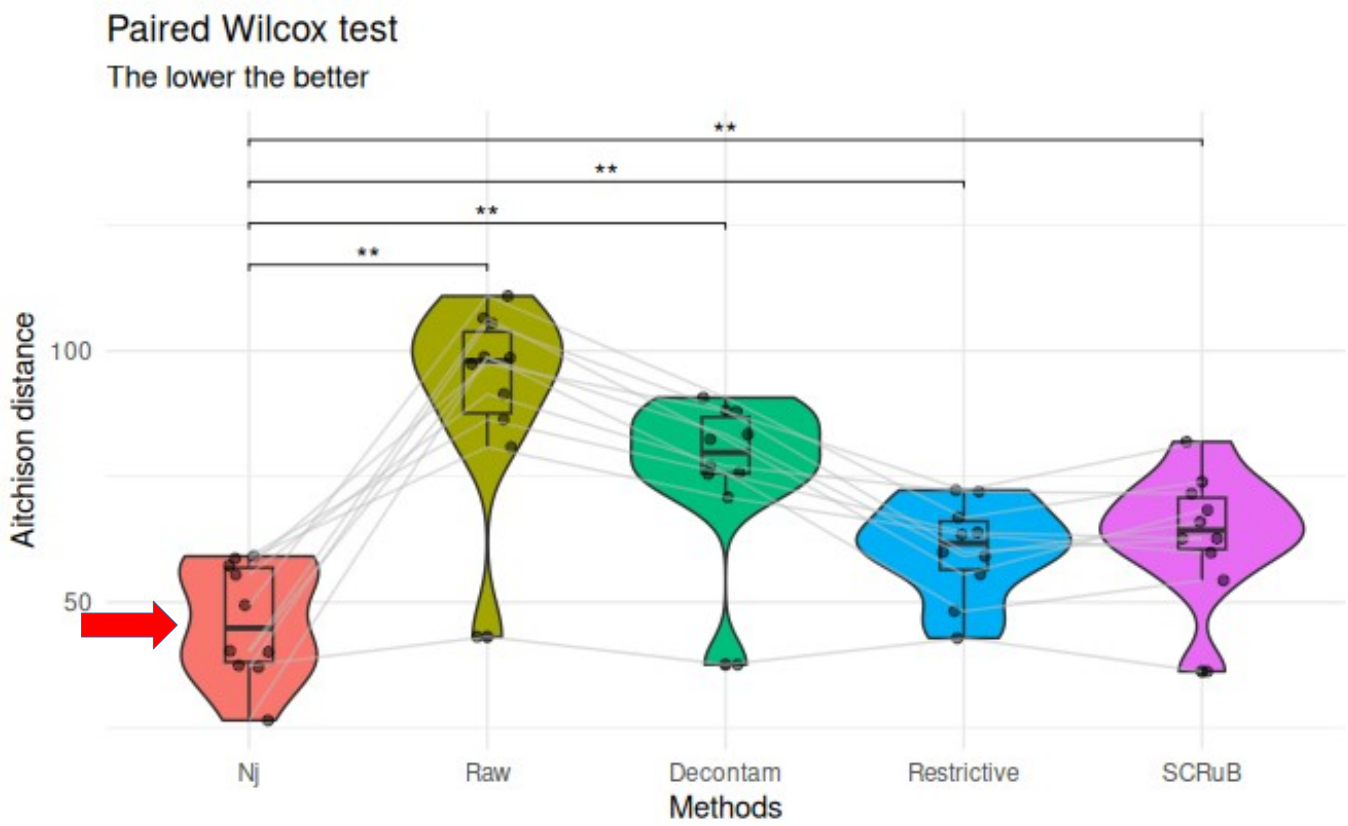
- Bacterial profile before/after decontaminated
- longitudinal collection of NCT samples

- **Higher is better**



Assessment of Decontamination Approach with Technical Replica



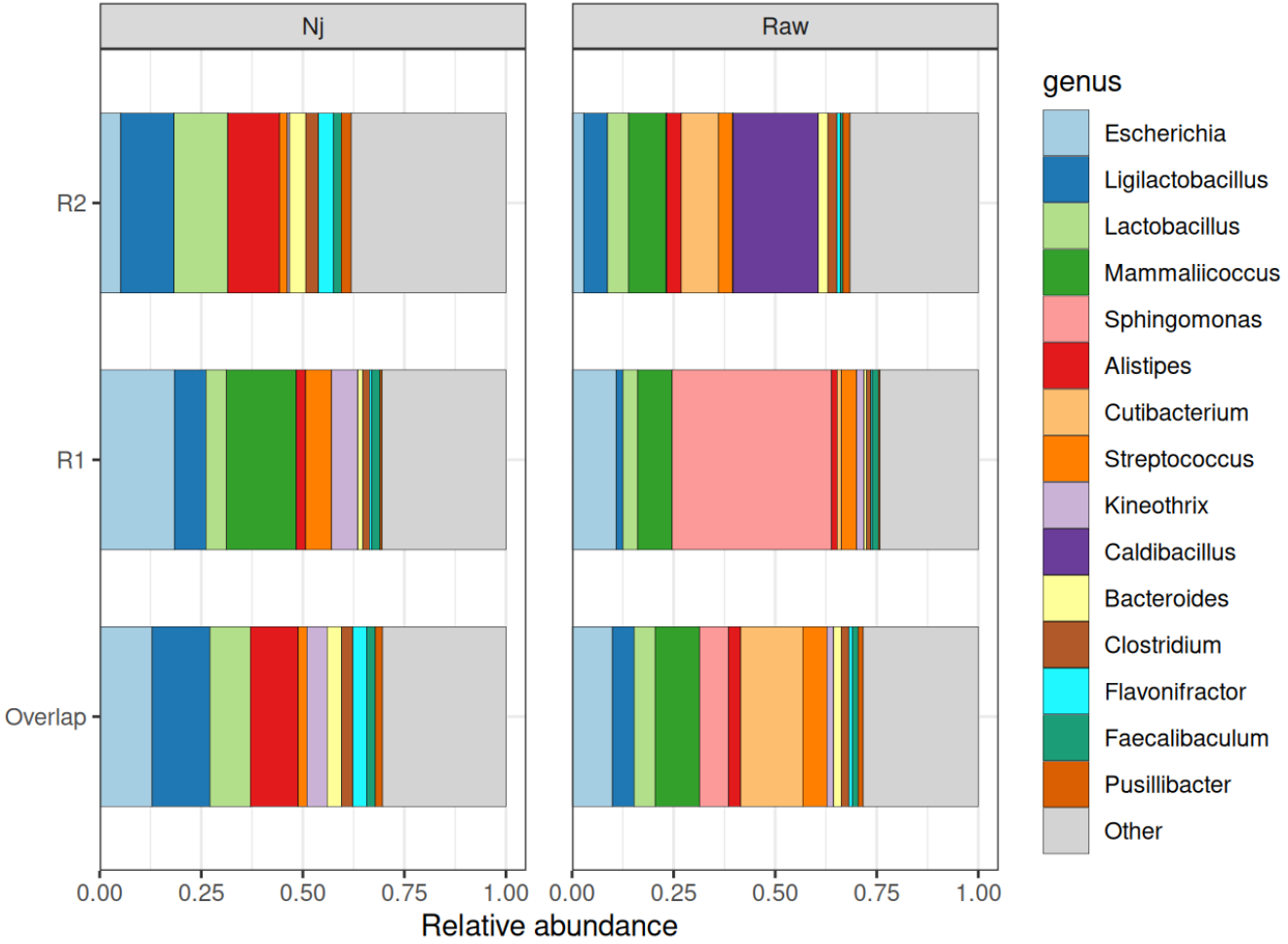


From 10 PDAC samples:

- Each has 2 replica.
- Distance between 2 replica before/after decontamination.
- **Lower is better**

Left: created by Biorender. Right: unpublished data

Intratumoral microbiome has been restored



Unpublished data

Summary

Negative Control Survey in the Lab

- NCT are essential to avoid spurious taxa.
- Beware of the batch effects.
- Thoroughly removing contaminant such as Nejman et al. procedure seems to outperforms others.

Intratumoral Microbiome in PDAC

- Intratumoral microbiota could be reliably derived from **technical replica**, **negative controls**, and **decontamination tools**.
- Important for further downstream analyses and machine learning.

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Thank you for your attention!

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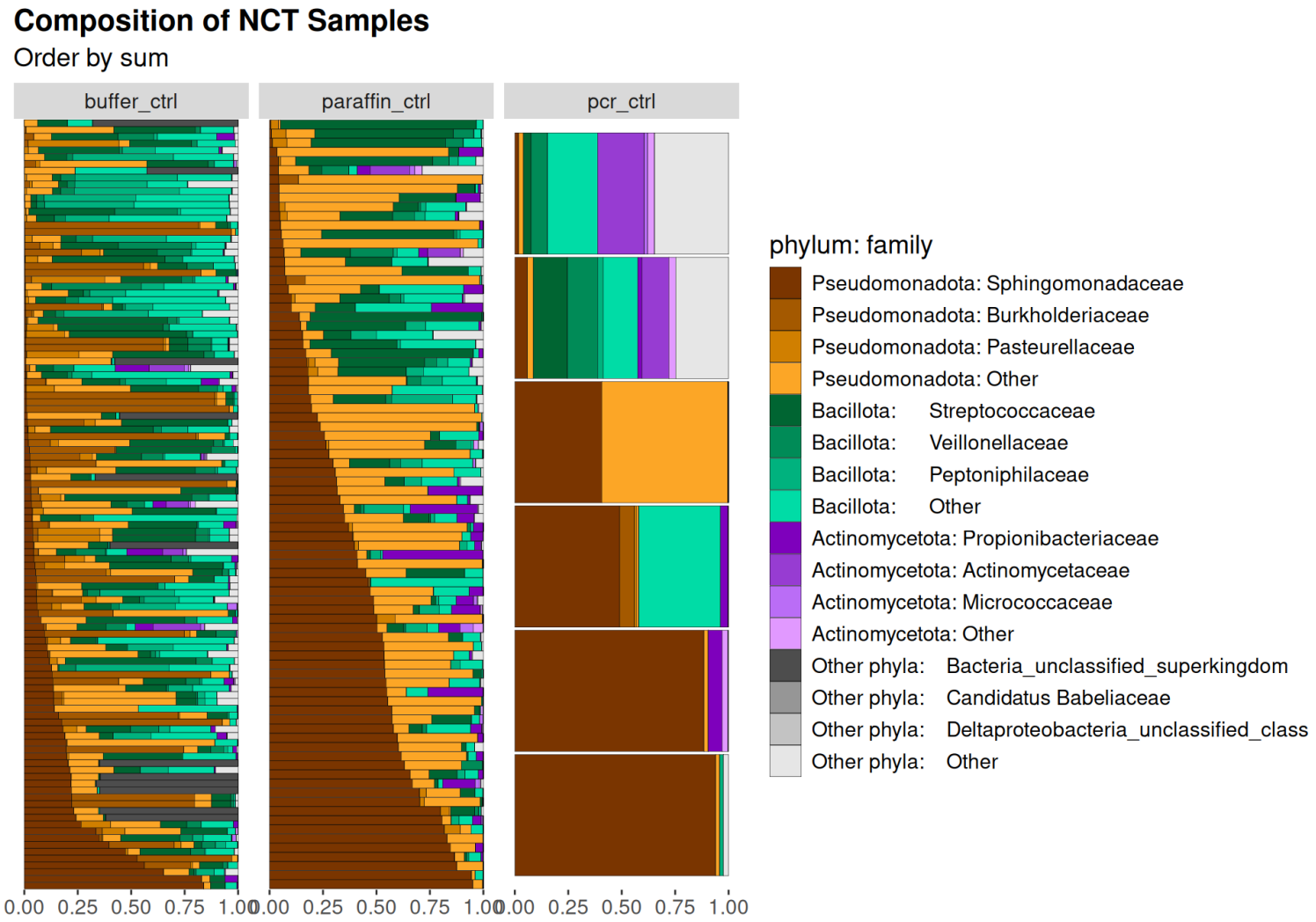
DFG Deutsche Forschungsgemeinschaft

train GI Cancer

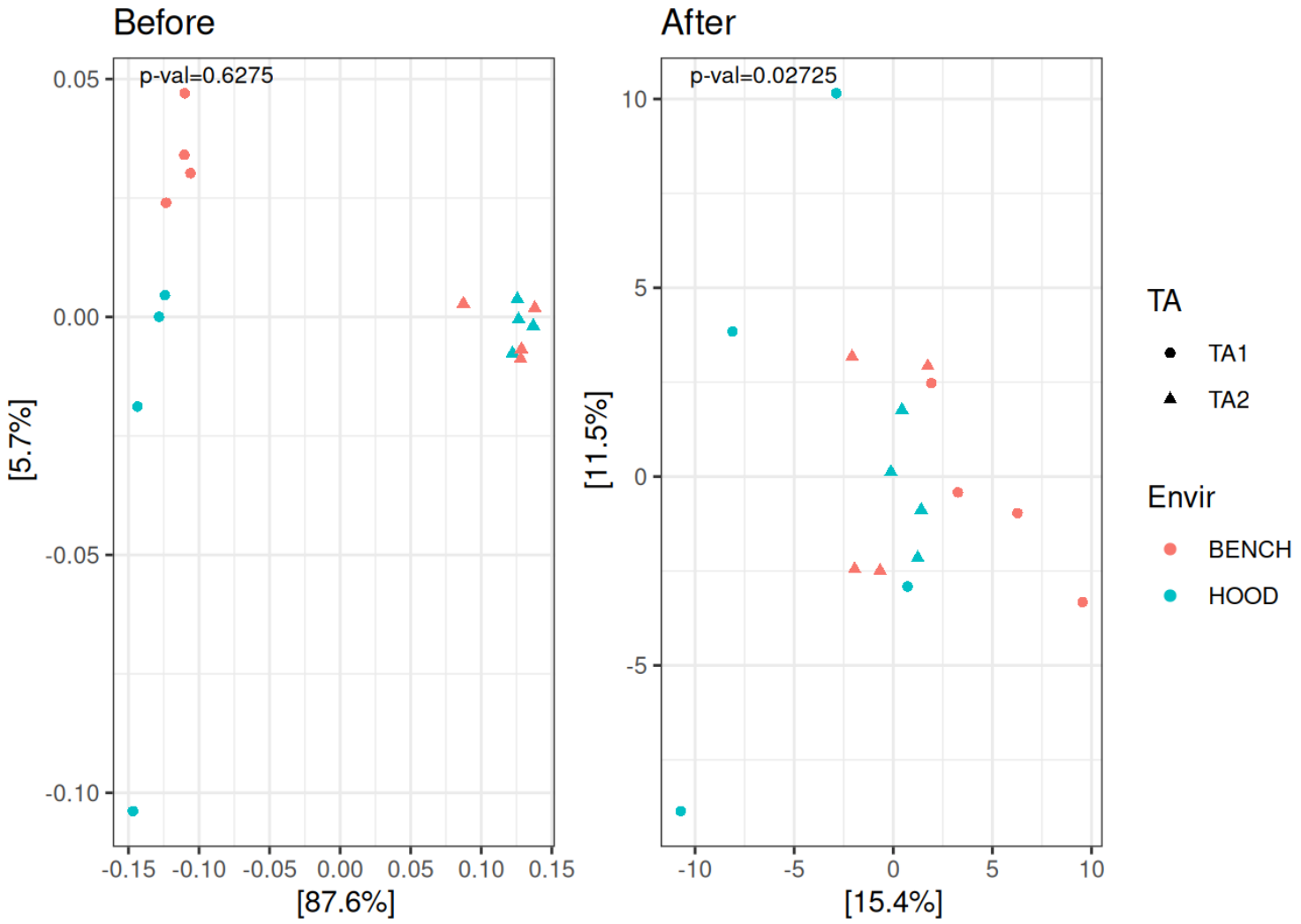
Backup

NCT – Bacterial Composition



Unpublished Data

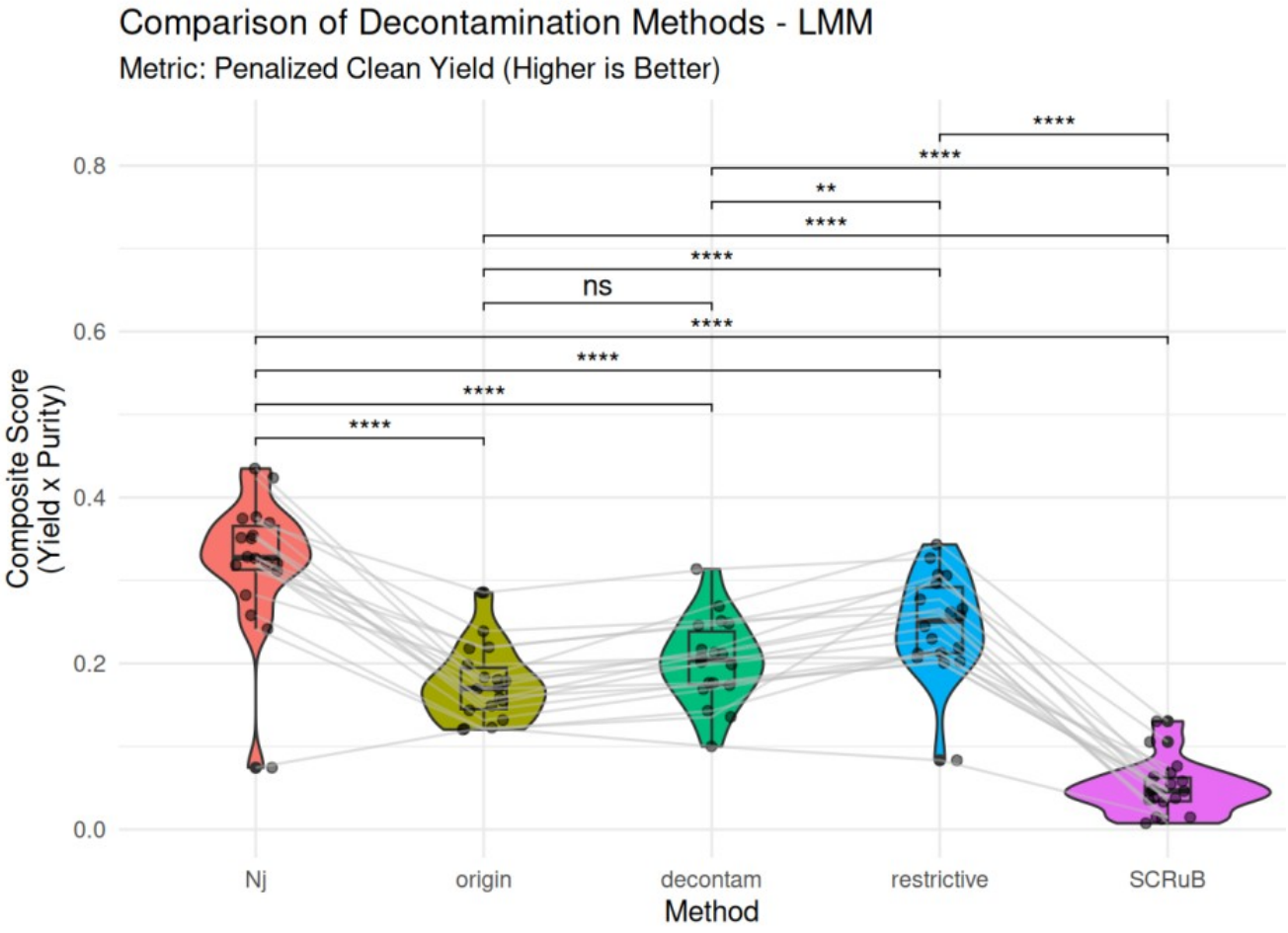
Before and after batch correction



Unpublished Data

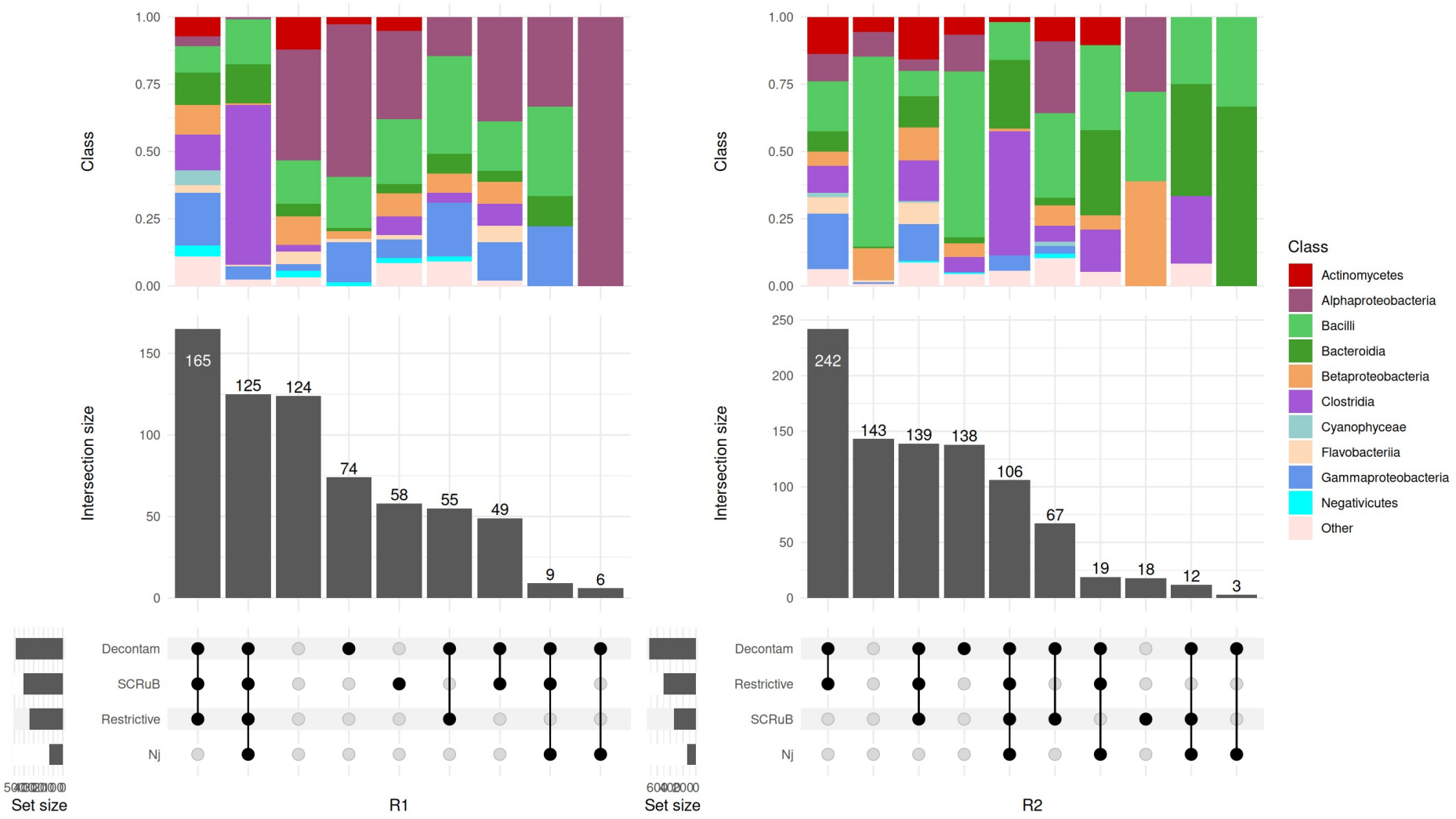
Composite Score Calculation

- **Yield**:
$$\frac{\#\ putative\ true\ taxa \notin NCT}{\#observed\ species}$$
- **Purity**:
$$\frac{\#\ putative\ true\ taxa \notin NCT}{\#putative\ contaminants}$$
- $Composite\ Score = Yield \times Purity$



Unpublished Data

Overlap among various decontamination methods



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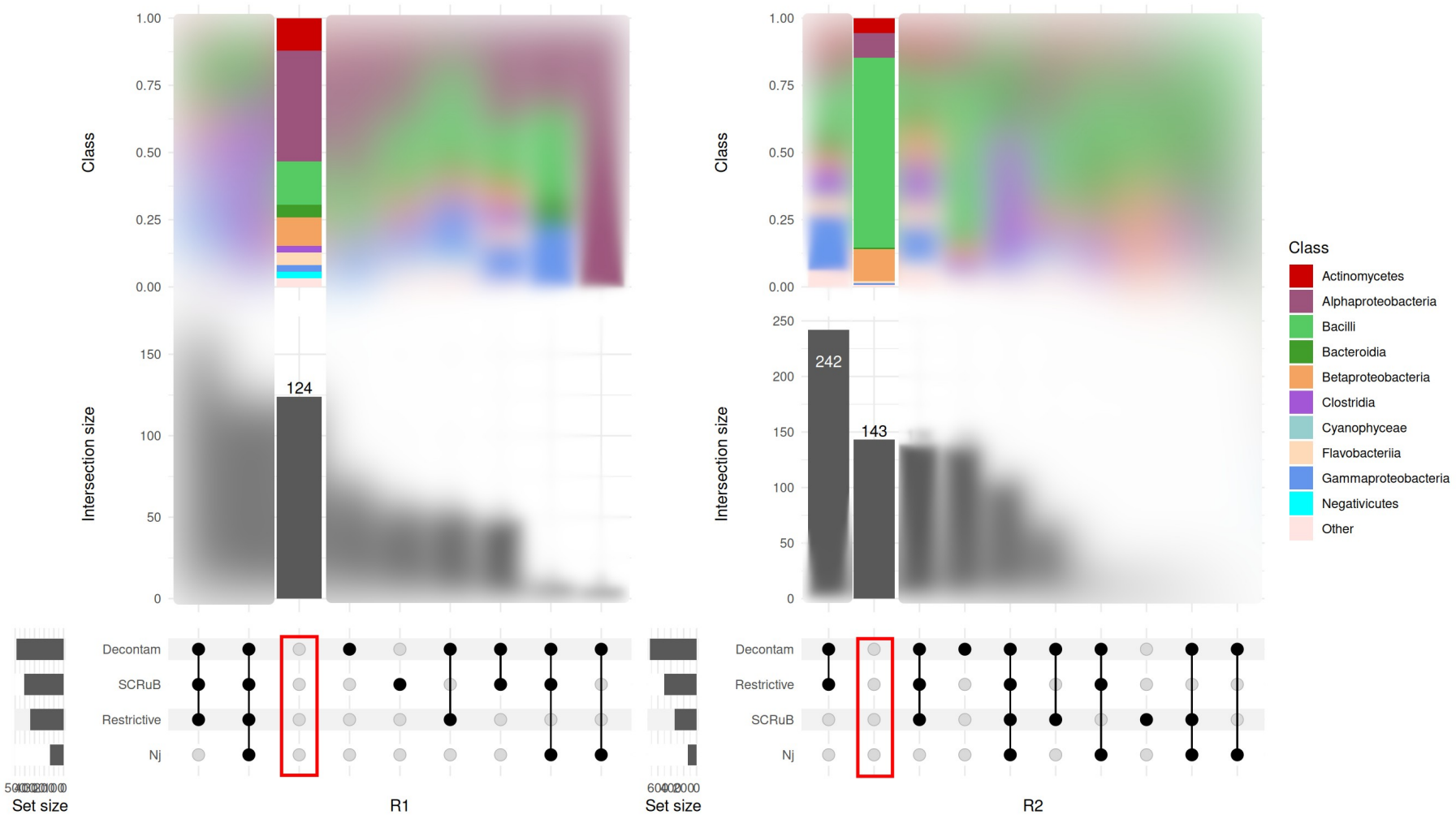

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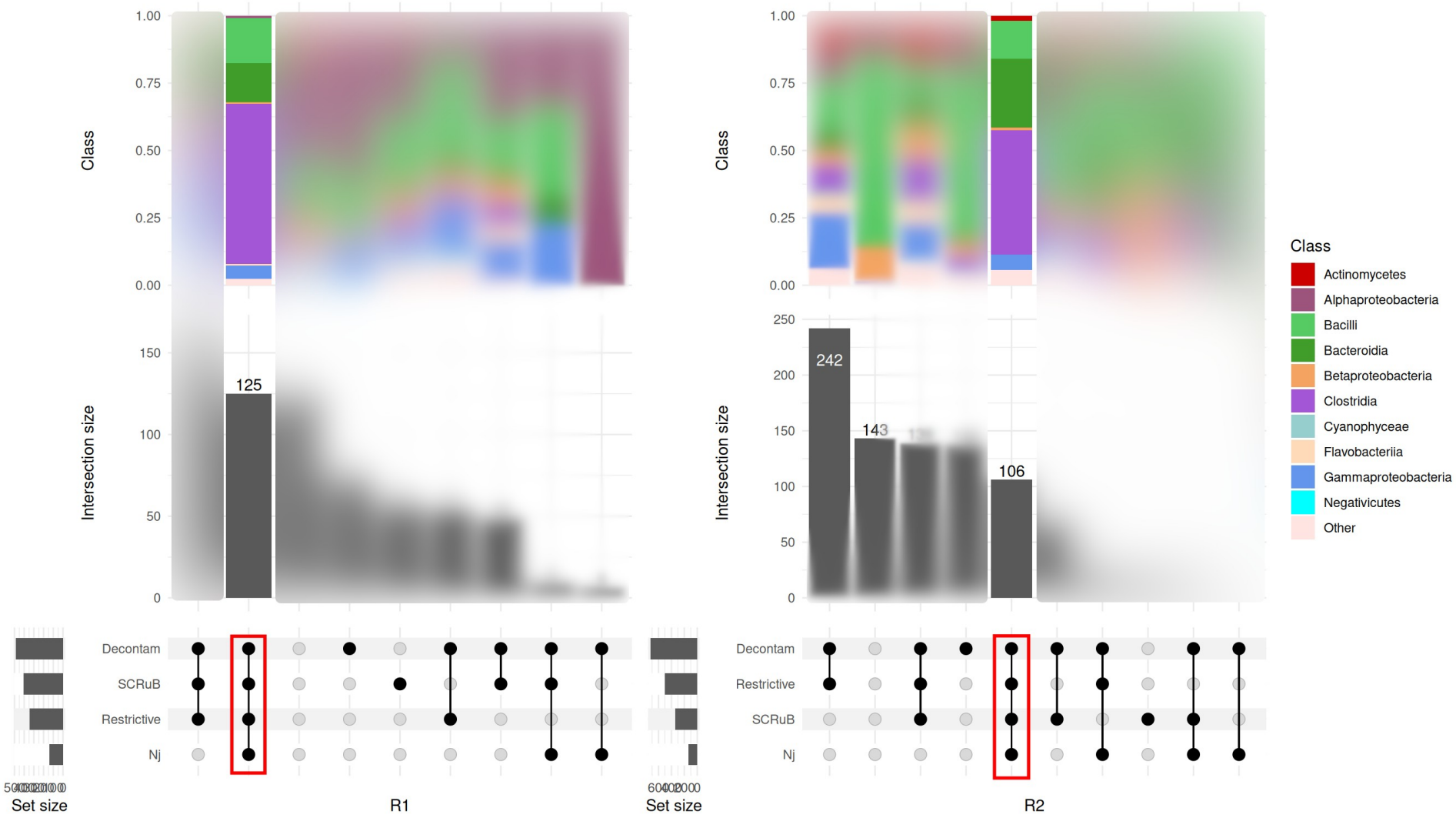

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